

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Full Stack Development Using Mern Stack

Practical – VIII

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Class: TYIT	Batch: 2
Date of Assignment: 19-09-26	Date/Time of Submission: 19-09-26

API Creation and Integration

a. Create a simple REST API using Express

```
const express = require("express");

const mongoose = require("mongoose");

const cors = require("cors");

const app = express();

const PORT = 3000; // Middleware

app.use(cors()); app.use(express.json()); // Connect MongoDB

mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {

console.log("MongoDB Connected Successfully"); })

.catch((error) => { console.log("MongoDB

Connection Failed");

console.log(error);

});
```

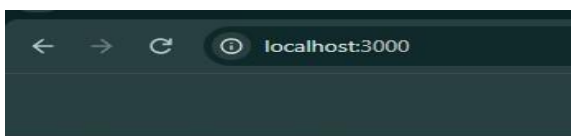
// Simple REST API

```
app.get("/", (req, res) => { res.send("Student Management API is

Running"); }); app.listen(PORT, () => { console.log(`Server is

running on http://localhost:${PORT}`); });
```

```
Node.js v24.17.0
PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical> cd Practical8.ipynb
PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\Practical8.ipynb> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
[]
```



Student Management API is Running

b. Implement a GET API to retrieve data.

```
const express = require("express");

const mongoose = require("mongoose");
```

```

const cors = require("cors");

const app = express();

const PORT = 3000; // Middleware

app.use(cors());

app.use(express.json()); // Connect MongoDB

mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {
  console.log("MongoDB Connected Successfully"); })
  .catch((error) => { console.log("MongoDB
  Connection Failed"); console.log(error);
  });

```

// Simple REST API

```

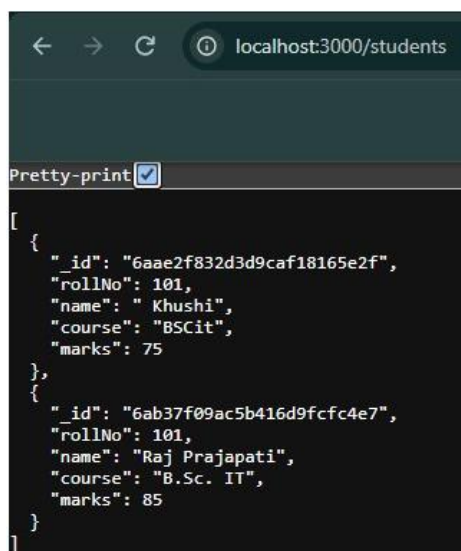
app.get("/", (req, res) => { res.send("Student
Management API is Running"); }); app.get("/students",
async (req, res) => { try {
  const students = await mongoose.connection.db
.collection("Students")
  .find() .toArray();
  res.json(students);
} catch (error) {
  res.status(500).json({ message:
  "Error fetching students"
  });
} }); app.listen(PORT, () => { console.log(`Server is running on
http://localhost:${PORT}`); });

```

```

PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\Practical8.ipynb> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully

```



```

[
  {
    "_id": "6aae2f832d3d9caf18165e2f",
    "rollNo": 101,
    "name": "Khushi",
    "course": "BSCit",
    "marks": 75
  },
  {
    "_id": "6ab37f09ac5b416d9fcfc4e7",
    "rollNo": 101,
    "name": "Raj Prajapati",
    "course": "B.Sc. IT",
    "marks": 85
  }
]

```

c. Fetch data from the API in a React application.

```
import { useEffect, useState } from "react";
import axios from "axios"; function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => {
    axios.get("http://localhost:3000/students")
      .then((response) => {      setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>  ) )}
        </tbody>
      </table>
    </div>  ); } export default App;
```

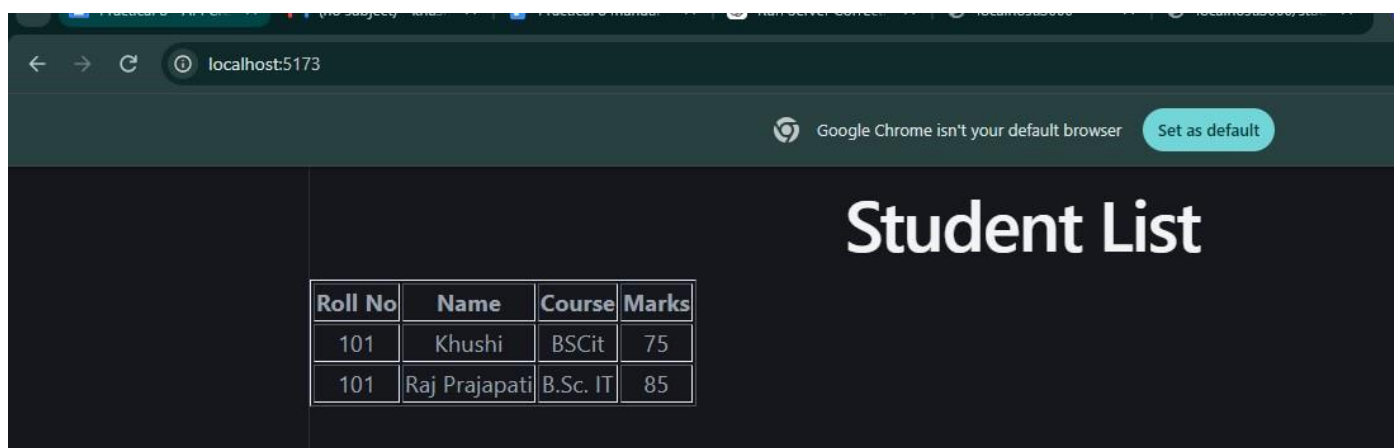
```

PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\my-react> npm run dev

→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
1:18:45 pm [vite] (client) hmr update /src/App.jsx
1:18:45 pm [vite] (client) hmr update /src/App.jsx (x2)
1:18:45 pm [vite] (client) dependency optimized: axios
1:18:45 pm [vite] (client) optimized dependencies changed
. reloading

PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\Practical8.ipynb> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully

```



d. Send data from React to the server using a POST request.

1) Replace App.jsx code

```

import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");

  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };

  useEffect(() => {
    getStudents();
  }, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios.post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
        setName("");
        setCourse("");
        setMarks("");
        getStudents();
      })
  };

```

```

.catch((error) => { console.log(error);
}); });
return (
  <div>
    <h1>Student Management System</h1>
    <h2>Add Student</h2>
    <input type="number" placeholder="Roll No"
value={rollNo} onChange={(e) =>
setRollNo(e.target.value)} />
    <br /><br /> <input type="text"
placeholder="Student Name" value={name}
onChange={(e) => setName(e.target.value)} />
    <br /><br /> <input type="text"
placeholder="Course" value={course}
onChange={(e) => setCourse(e.target.value)} />
    <br /><br /> <input type="number"
placeholder="Marks" value={marks}
onChange={(e) => setMarks(e.target.value)} />
    <br /><br />
    <button onClick={addStudent}>
      Add Student
    </button>
    <hr />
    <h2>Student List</h2>
    <table border="1">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>
      <tbody>
        {students.map((student) => (
          <tr key={student._id}>

```

```

<td>{student.rollNo}</td> <td>{student.name}</td>
<td>{student.course}</td> <td>{student.marks}</td>
</tr>  )}}
</tbody>
</table>
</div> );} export default App;

```

2)Add post method in server.js file above app.listen()

```

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000; // Middleware
app.use(cors());
app.use(express.json()); // Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {
console.log("MongoDB Connected Successfully"); })
.catch((error) => { console.log("MongoDB
Connection Failed"); console.log(error);
});

```

// Simple REST API

```

app.get("/", (req, res) => { res.send("Student
Management API is Running"); }); app.get("/students",
async (req, res) => { try {
const students = await mongoose.connection.db
.collection("Students")
.find() .toArray();
res.json(students);}
catch (error) { res.status(500).json({
message: "Error fetching students"});
}); app.post("/students", async (req,
res) => { try { const student =
req.body; await
mongoose.connection.db
.collection("students")
.insertOne(student); res.json({ message:

```

"Student added successfully",

student:student

```
}); } catch (error) {
```

```
res.status(500).json({ message:
```

```
"Error adding student" }); } });
```

```
app.listen(PORT, () => { console.log(`Server is running on
```

```
http://localhost:${PORT}`); });
```

3) Now 1st execute server.js then app.jsx in different terminals.

```

PS C:\Users\info\OneDrive\Desktop\mern Stack Practical\my
-learn> npm run dev

> my-react@0.0.0 dev
> vite

VITE v8.3.0 ready in 276 ms
  ➔ Local:   http://localhost:5173/
  ➔ Network: use --host to expose
  ➔ press h + enter to show help

PS C:\Users\info\OneDrive\Desktop\mern Stack Practical\Practic
al8.ipynb>

...
al8.ipynb> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
  
```

